

Project 1

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Map

```
#Ohio map
map.county <- map_data('county')
ohio.county <- subset(map.county, region=="ohio")

#get corona cases from ohio.gov
ohioDF <- read_csv("https://coronavirus.ohio.gov/static/COVIDSummaryData.csv")
#get ohio population from census.gov source = https://www.census.gov/data/datasets/time-
#series/demo/popest/2010s-counties-total.html#par_textimage_739801612
ohioPop <- read_excel("co-est2019-annres-39.xlsx")

#get the date
Today <- Sys.Date()

#filter out row of totals and mutate variables
ohioDF <- ohioDF %>%
  filter(Sex != "Total") %>%
  mutate(OnsetDate = mdy('Onset Date'))
#filter(Today - OnsetDate <= 30) %>% -----no more filter by time occ

#Creates dataset grouped by County with case, hospitalization, and death count, used for map and for bo
ohioCountyDF <- ohioDF %>%
  group_by(County) %>%
  summarize(ncases = sum('Case Count'),
            ndead = sum('Death Due to Illness Count', na.rm = TRUE),
            nhos = sum('Hospitalized Count', na.rm = TRUE)) %>%
  mutate(CaseCat = cut(ncases,
                       breaks = c(0, 1000, 2500, 5000,
                                   15000, 30000, 50000))) %>%
  mutate(HosCat = cut(nhos,
                     breaks = c(0, 100, 500,
                                   1000, 2000, 2500, 3000))) %>%
  mutate(DeathCat = cut(ndead,
                       breaks = c(0, 150,
                                   200, 350, 600, 700)))

#filter out unneeded rows and columns before combining tables
ohioPop <- ohioPop[c(-1, -2, -3, -4, -93, -94, -95, -96, -97),2]
```

```

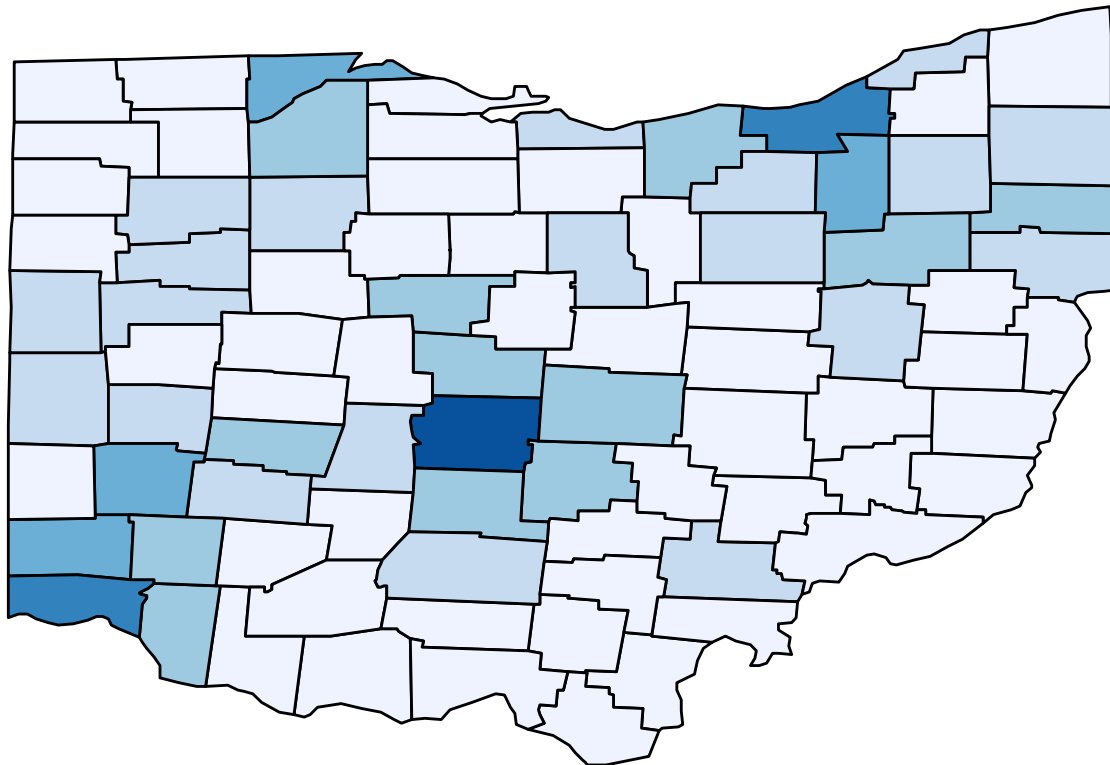
#combine tables and changing variables
#rename county to region to map it with ohio.county
#https://www.sharpsightlabs.com/blog/rename-columns-in-r/
combinedOhio <- cbind(ohioCountyDF, ohioPop) %>%
  mutate(countyPop = as.double(...2)) %>%
  select(!...2) %>%
  mutate(rate = (ncases/countyPop) * 1000) %>%
  mutate(County = str_to_lower(County)) %>%
  rename(subregion = County)

#combine all data sets into one for state map
rateDB <- merge(ohio.county, combinedOhio, by.x = "subregion", by.y = "subregion")

#ohio map with continuous shading by total cases //Need to be categorical?
ohioMap <- ggplot(data = rateDB, aes(x=long,y=lat,group = group, fill=CaseCat)) +
  geom_polygon(color = "black") +
  scale_fill_brewer() +
  guides(fill = FALSE) +
  theme_map()

ohioMap

```

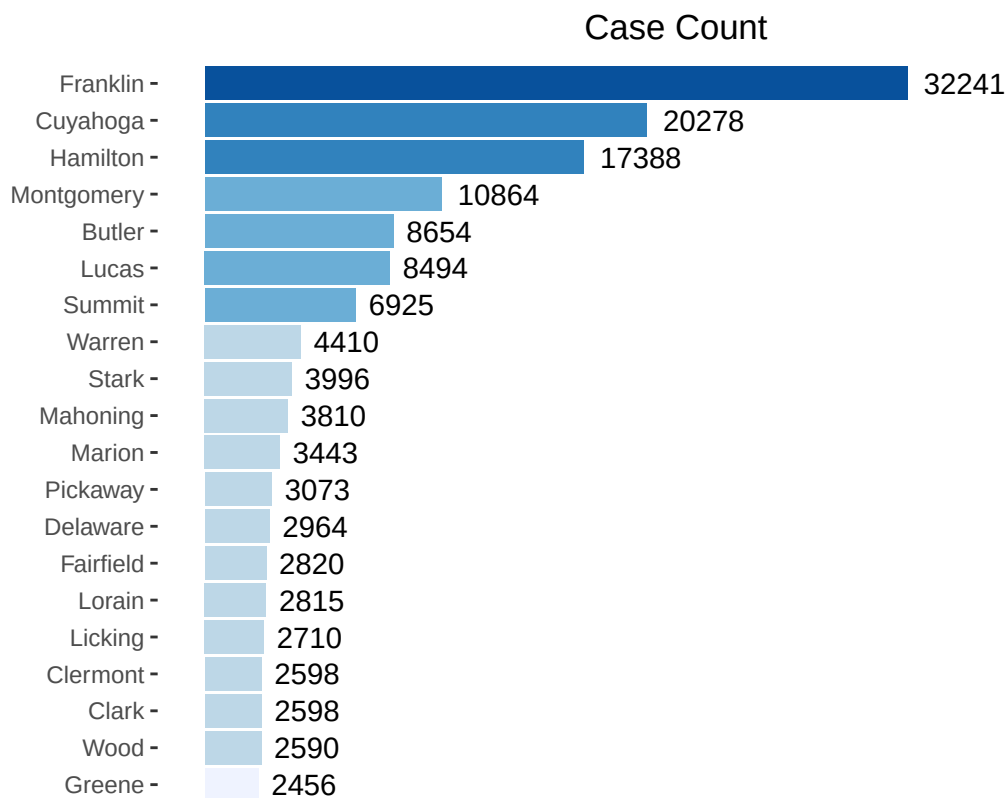


Bottom panels

#Bottom left case graph which displays the top 20 counties by cases

```
Cases <- ggplot(data = ohioCountyDF[tail(order(ohioCountyDF$ncases), 20), ], aes(y=reorder(County, ncases),
  geom_col() +
  scale_fill_brewer() +
  guides(fill = FALSE) +
  geom_text(aes(label = ncases), hjust = -.2) +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank(), plot.title = element_text(hjust = 0.5)) +
  xlim(0, max(ohioCountyDF$ncases + 11000)) +
  labs(y = "", x = "", title = "Case Count")
```

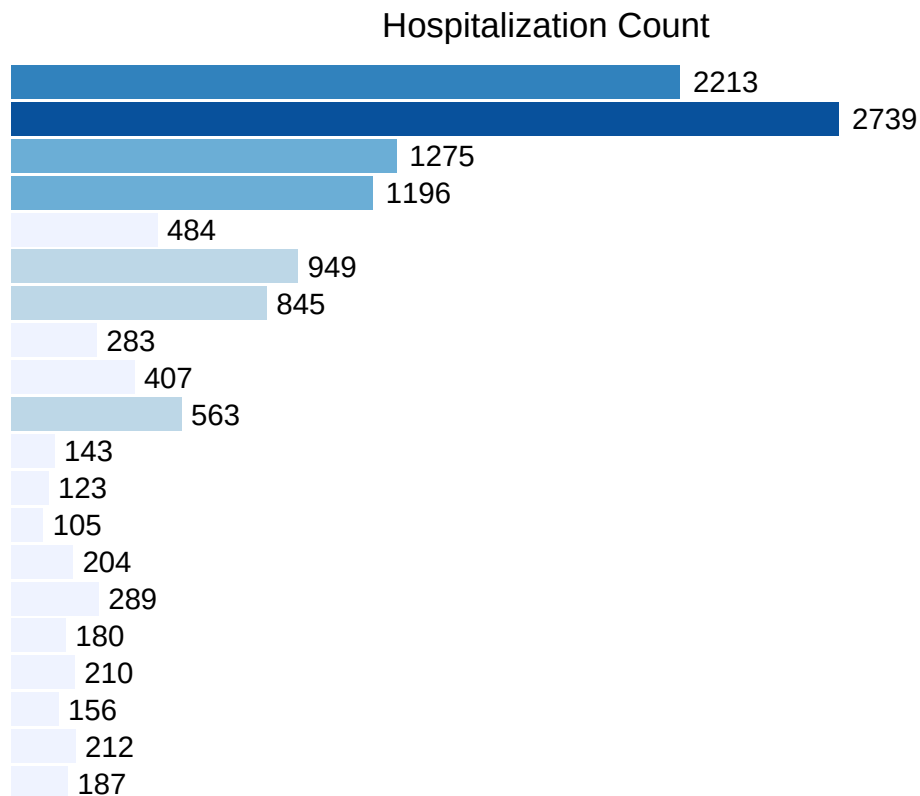
Cases



#Bottom middle hospitalization graph which displays the top 20 counties by cases

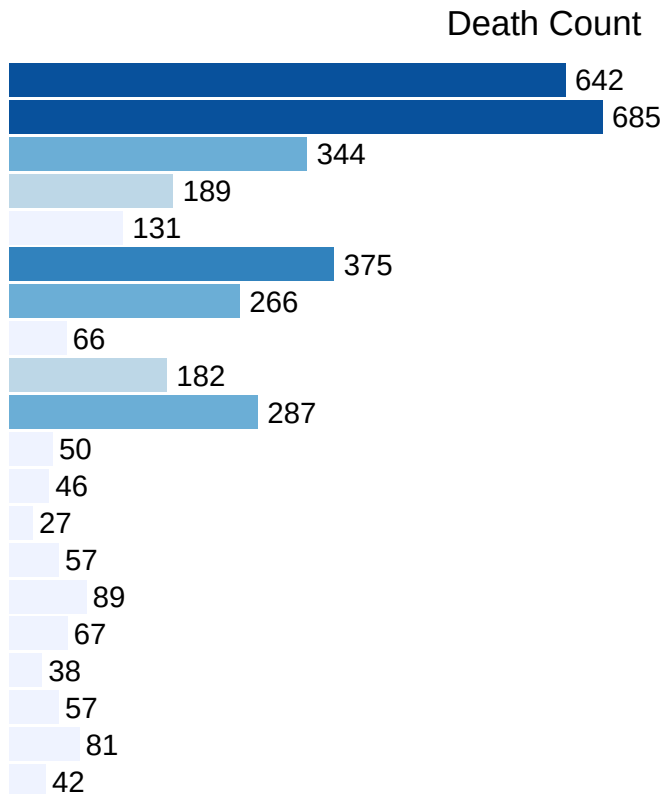
```
Hosp <- ggplot(data = ohioCountyDF[tail(order(ohioCountyDF$ncases), 20), ], aes(y=reorder(County, ncases),
  geom_col() +
  scale_fill_brewer() +
  guides(fill = FALSE) +
  geom_text(aes(label = nhos), hjust = -.2) +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank(), plot.title = element_text(hjust = 0.5)) +
  xlim(0, max(ohioCountyDF$nhos + 800)) +
  labs(y = "", x = "", title = "Hospitalization Count")
```

Hosp



#Bottom right death graph which displays the top 20 counties by cases

```
Deaths <- ggplot(data = ohioCountyDF[tail(order(ohioCountyDF$ncases), 20), ], aes(y=reorder(County, nca
  geom_col() +
  scale_fill_brewer() +
  guides(fill = FALSE) +
  geom_text(aes(label = ndead), hjust = -.2) +
  theme(axis.text = element_blank(), axis.ticks = element_blank(), plot.title = element_text(hjust = 0.
  xlim(0, max(ohioCountyDF$ndead + 550)) +
  labs(y = "", x = "", title = "Death Count")
Deaths
```



```
#combine all three graphs into one object
bottom <- Cases + Hosp + Deaths
```

Side Panels

```
#Dataset for right panel of four charts
ohioCasesDF <- ohioDF %>%
  group_by(OnsetDate) %>%
  summarize(ncases = sum('Case Count'),
            ndead = sum('Death Due to Illness Count', na.rm = TRUE),
            nhos = sum('Hospitalized Count', na.rm = TRUE))

#breaks for dates on x-axis graph
datebreaks <- seq(as.Date("2020-01-01"), as.Date("2020-11-01"), by = "2 months")

#right top case graph
casesBar <- ggplot(data = ohioCasesDF, aes(x = OnsetDate, y = ncases)) +
  geom_col(color = "blue") +
  scale_x_date(breaks = datebreaks, labels = date_format("%b %d")) +
  scale_fill_brewer() +
  theme(axis.text.y = element_blank(), axis.ticks.y = element_blank(), plot.title = element_text(hjust = 1)) +
  labs(y = "", x = "", title = "Cases", subtitle = paste(sum(ohioCasesDF$ncases)))
```

```

#right upper middle hospitalization graph
hosBar <- ggplot(data = ohioCasesDF, aes(x = OnsetDate, y = nhos)) +
  geom_col(color = "blue") +
  scale_x_date(breaks = datebreaks, labels = date_format("%b %d")) +
  scale_fill_brewer() +
  theme(axis.text.y = element_blank(), axis.ticks.y = element_blank(), plot.title = element_text(hjust = 0.5),
  labs(y = "", x = "", title = "Hospitalizations", subtitle = paste(sum(ohioCasesDF$nhos)))

#right lower middle death graph
deadBar <- ggplot(data = ohioCasesDF, aes(x = OnsetDate, y = ndead)) +
  geom_col(color = "blue") +
  scale_x_date(breaks = datebreaks, labels = date_format("%b %d")) +
  scale_fill_brewer() +
  theme(axis.text.y = element_blank(), axis.ticks.y = element_blank(), plot.title = element_text(hjust = 0.5),
  labs(y = "", x = "", title = "Deaths", subtitle = paste(sum(ohioCasesDF$ndead)))

Today <- Sys.Date()

ohioRec <- ohioCasesDF %>%
  filter(OnsetDate < Today - 10) %>%
  mutate(recovered = ncases - ndead)

#right bottom graph for presumed recoveries
recBar <- ggplot(data = ohioRec, aes(x = OnsetDate + 10, y = recovered)) +
  geom_col(color = "blue") +
  scale_x_date(breaks = datebreaks, labels = date_format("%b %d")) +
  scale_fill_brewer() +
  theme(axis.text.y = element_blank(), axis.ticks.y = element_blank(), plot.title = element_text(hjust = 0.5),
  labs(y = "", x = "", title = "Presumed Recovered", subtitle = paste(sum(ohioRec$recovered)))

right <- casesBar / hosBar / deadBar / recBar

```

Pulling it all together

```

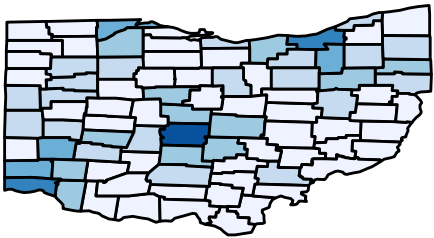
(ohioMap + right) / bottom

```

Hospitalizations

18095

Jan 01 Mar 01 May 01 Jul 01 Sep 01 Nov 01



Deaths

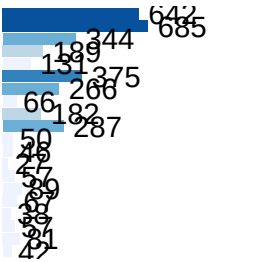
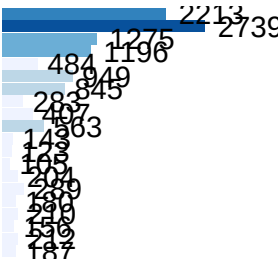
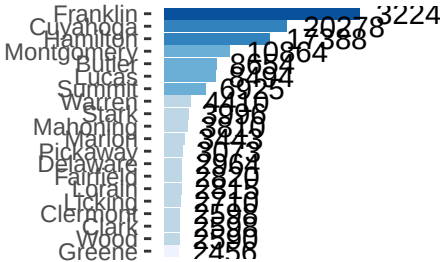
5206

Jan 01 Mar 01 May 01 Jul 01 Sep 01 Nov 01

Case Count

Hospitalization Count

Death Count



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